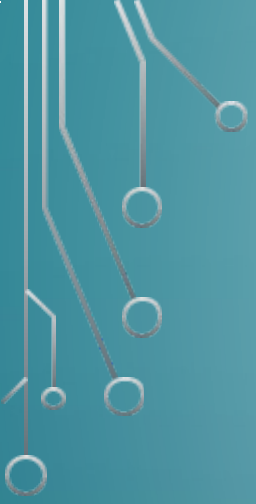


A decorative graphic on the left side of the slide, consisting of a network of thin, light-colored lines and small circles, resembling a circuit board or a stylized tree structure, set against a teal-to-blue gradient background.

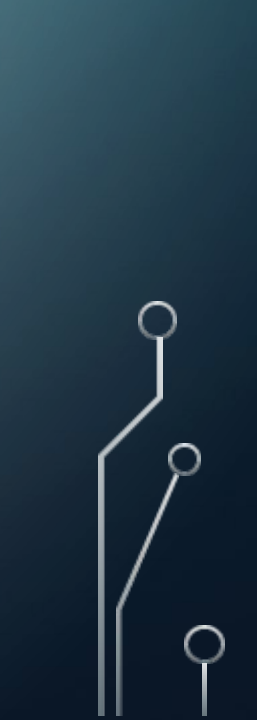
# ARM PROCESSOR AND APPLICATION

COURSE PROJECT



**Tilte:-**

Temperature-Controlled Stepper  
Motor System for Automated  
Cooling

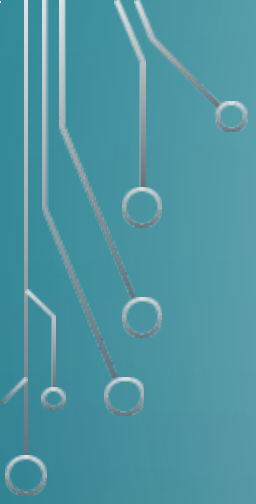


TEAM NO:8

| NAME    | RNO | USN          |
|---------|-----|--------------|
| Sakshi  | 636 | 01FE23BEC335 |
| Shivani | 638 | 01FE23BEC337 |
| Arpita  | 656 | 01FE23BEC357 |
| Sneha   | 622 | 01FE23BEC321 |

# INTRODUCTION

The **Temperature-Controlled Stepper Motor System for Automated Cooling** is an embedded project developed using the **ARM7 LPC2148** microcontroller and **Keil IDE**. It aims to automatically regulate cooling based on ambient temperature. A temperature sensor (e.g., LM35) continuously monitors the environment, and the sensor's analog data is processed using the LPC2148's inbuilt ADC. When the temperature exceeds a defined threshold, the microcontroller activates a **stepper motor** to control a cooling mechanism such as vents or fans. This system offers precise control, energy efficiency, and real-time response, making it ideal for smart cooling applications in industrial and residential settings.



## System Behavior:

When temperature rises (heating) → Stepper motor stops.

When temperature falls (cooling) → Stepper motor starts/rotates.





# CODE

```
1#include <lpc214x.h>
2#include <stdio.h>
3unsigned int var1,k;
4unsigned int i;
5void delay(unsigned long int rl)
6{
7    volatile unsigned long int r;
8    for (r = 0; r < rl*20; r++);
9}
10
11unsigned int adc_value = 0, temp_adc = 0;
12char val[30];
13
14void serial()
15{
16    PINSEL0 = 0x00000005;
17    UOLCR = 0x83;
18    UODLM = 0x00;
19    UODLL = 0x61;
20    UOLCR = 0x03;
21}
22
23void clock_wise(void)
24{
25    var1 = 0x00000800;           //For Clockwise
26    for( i = 0 ; i <= 3 ; i++ )   // for A B C D Stepping
27    {
28        var1 <<= 1;              //For Clockwise
29
30        IOOCLR =0x0000F000 ;      //clearing all 4 bits
31
32        IOOSET = var1 ;           // setting perticular bit
33}
```

## Output

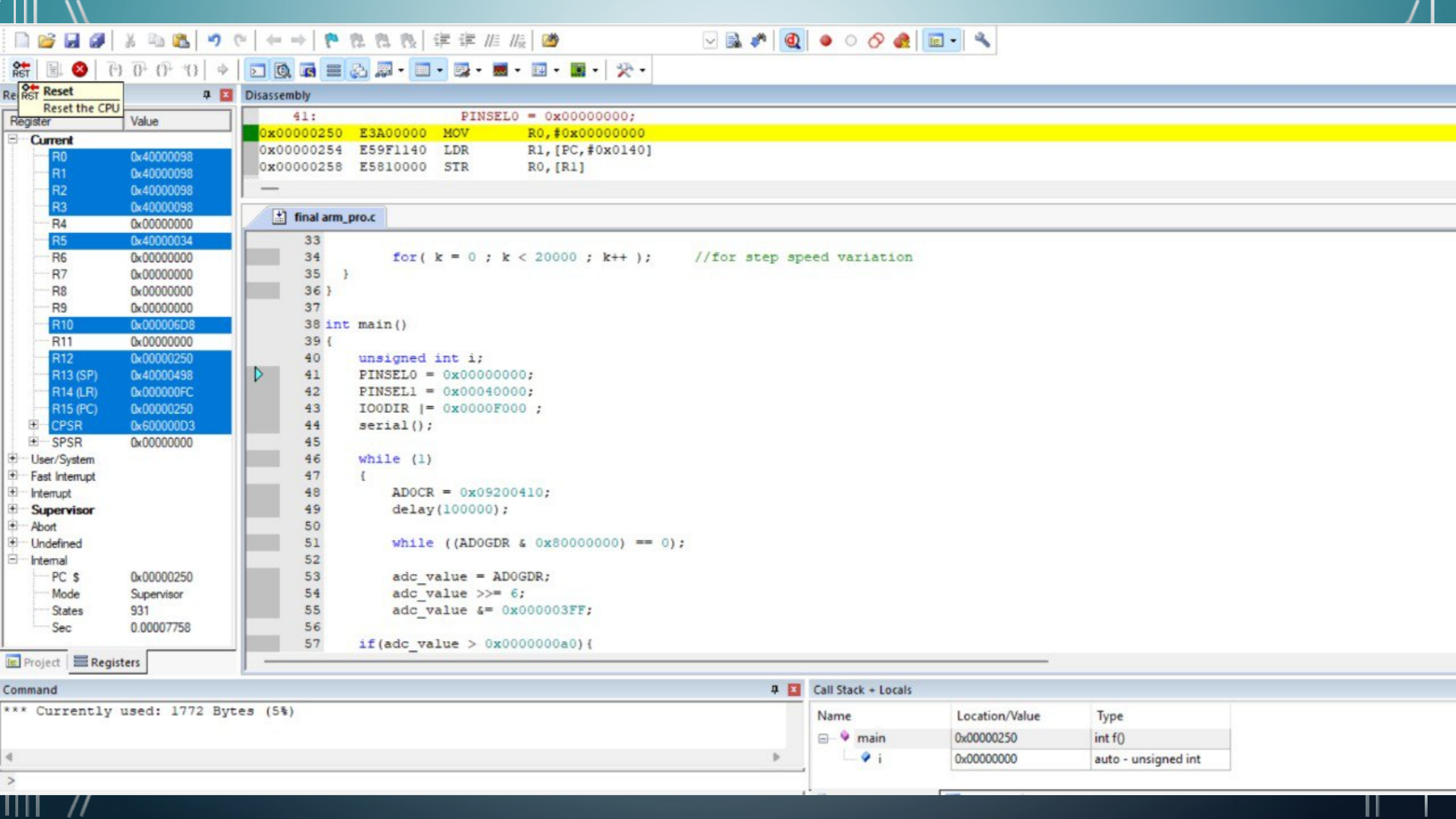
```
Build target 'Target 1'
Assembling Startup.s...
Compiling final_arm_pro.c...
final_arm_pro.c(77): warning: #1-D: last line of file ends without a newline
```

```
final_arm_pro.c: 1 warning, 0 errors
Linking...
```

```
Program Size: Code=1680 RO-data=72 RW-data=20 ZI-data=1292
```

```
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Target 1
Project
Target 1
Source Group 1
Startup.s
final_arm_pro.c

37
38 int main()
39 {
40     unsigned int i;
41     PINSEL0 = 0x00000000;
42     PINSEL1 = 0x00040000;
43     IOODIR |= 0x0000F000 ;
44     serial();
45
46     while (1)
47     {
48         AD0CR = 0x09200410;
49         delay(1000000);
50
51         while ((AD0GDR & 0x80000000) == 0);
52
53         adc_value = AD0GDR;
54         adc_value >>= 6;
55         adc_value &= 0x000003FF;
56
57         if(adc_value > 0x0000000a0){
58             sprintf(val, "OVERHEAT !! %x", adc_value);
59             IOOCLR = 0x0000F000;
60         }
61         else{
62             sprintf(val, "UNDER CONTROL !! %x", adc_value);
63             for(i=0;i<150;i++)
64                 clock_wise();
65         }
66         while (val[i] != '\0')
67         {
68             while (!(UOLSR & 0x20));
69             UOTHR = val[i];
70             i++;
71         }
72
73         delay(500000);
74         UOTHR = '\t';
75         i = 0;
76     }
77 }
```



| Register       | Value      |
|----------------|------------|
| Current        |            |
| R0             | 0x40000098 |
| R1             | 0x40000098 |
| R2             | 0x40000098 |
| R3             | 0x40000098 |
| R4             | 0x00000000 |
| R5             | 0x40000034 |
| R6             | 0x00000000 |
| R7             | 0x00000000 |
| R8             | 0x00000000 |
| R9             | 0x00000000 |
| R10            | 0x000006D8 |
| R11            | 0x00000000 |
| R12            | 0x00000250 |
| R13 (SP)       | 0x40000498 |
| R14 (LR)       | 0x000000FC |
| R15 (PC)       | 0x00000250 |
| CPSR           | 0x600000D3 |
| SPSR           | 0x00000000 |
| User/System    |            |
| Fast Interrupt |            |
| Interrupt      |            |
| Supervisor     |            |
| Abort          |            |
| Undefined      |            |
| Internal       |            |
| PC \$          | 0x00000250 |
| Mode           | Supervisor |
| States         | 931        |
| Sec            | 0.00007758 |

```
41: PINSEL0 = 0x00000000;
0x00000250 E3A00000 MOV R0,#0x00000000
0x00000254 E59F1140 LDR R1,[PC,#0x0140]
0x00000258 E5810000 STR R0,[R1]
```

```
final arm_pro.c
33
34     for( k = 0 ; k < 20000 ; k++ );    //for step speed variation
35 }
36 }
37
38 int main()
39 {
40     unsigned int i;
41     PINSEL0 = 0x00000000;
42     PINSEL1 = 0x00040000;
43     IOODIR |= 0x0000F000 ;
44     serial();
45
46     while (1)
47     {
48         ADCR = 0x09200410;
49         delay(100000);
50
51         while ((AD0GDR & 0x80000000) == 0);
52
53         adc_value = AD0GDR;
54         adc_value >>= 6;
55         adc_value &= 0x000003FF;
56
57         if(adc_value > 0x0000000a0){
```

Command

\*\*\* Currently used: 1772 Bytes (5%)

| Name | Location/Value | Type                |
|------|----------------|---------------------|
| main | 0x00000250     | int f()             |
| i    | 0x00000000     | auto - unsigned int |

## OUTPUT VIDEO



## OUTPUT PHOTO



## CONCLUSION:

If the temperature rises above the threshold, the controller halts the stepper motor to prevent thermal buildup. Once the temperature drops below the threshold, the controller restarts the motor, allowing motion only under safe thermal conditions. This closed-loop feedback system ensures automated cooling through controlled actuation, enhancing safety and efficiency.